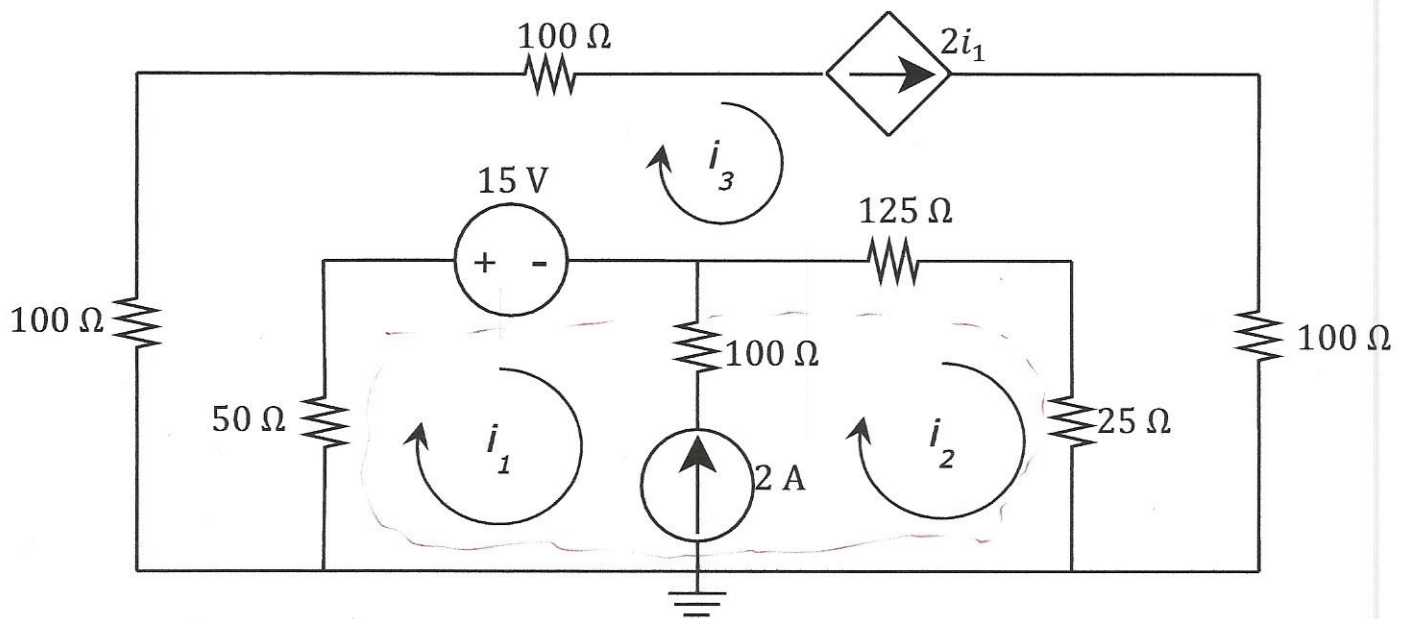


3) Use mesh analysis to set up two equations of the following form:

$$ai_1 + bi_2 = -3 A$$

$$ci_1 + di_2 = -2 A$$



$$i_2 - i_1 = 2 \Rightarrow i_1 - i_2 = -2$$

$$\boxed{c = 1} \quad \boxed{d = -1}$$

$$SM: 50(i_1 - i_3) + 15 + 125(i_2 - i_3) + 25(i_2 - i_3) = 0$$

$$50i_1 + 150i_2 - 200i_3 = -15$$

$$\text{but } i_3 = 2i_1$$

$$50i_1 + 150i_2 - 400i_1 = -15$$

$$-350i_1 + 150i_2 = -15$$

$$-70i_1 + 30i_2 = -3$$

$$\boxed{a = -70} \quad \boxed{b = 30}$$